TOPIC 6 • CLIMATE RESILIENCE & ESSENTIAL SERVICES



RippleReady

Solve The Problem Before It Spreads

Autonomous disaster cascade interception & hyper-local mutual aid engine, protecting schools, frail care, and FNB Speedpoint merchants before small disruptions trigger community catastrophes.

PILOT JURISDICTION

Ward 18, George

Western Cape

CASCADE COMPUTATION

< 3 Milliseconds

Zero Hallucination Math

FINANCIAL INTEGRATION

FNB Instant Pay

Micro-Fuel Stipends

THE PROBLEM

Disasters Do Not Exist in Isolation

A single water main rupture on York Street triggers a 120-minute cascading chain reaction.

TIER 1: LIFE & CARE

Buffer: 2 Hours

Little Stars Crèche

40 toddlers run out of drinking water and sanitation. Without immediate reserves, the school must shut down immediately.

Impact: 40 Children Displaced

TIER 1: CRITICAL CARE

Buffer: 1 Hour

Rosemoor Fragile & Elderly

28 frail care residents require sterile drinking water and continuous medical sanitation for chronic care and hydration.

Impact: High Medical Risk

TIER 3: FNB MERCHANT

Buffer: 3 Hours

Mama K's Spaza & Corner Eatery

120 daily lunch patrons. Water outage halts food prep, leading to food spoilage, lost POS turnover, and SME loan default risk.

Impact: R4,800 Daily Turnover Loss

The Status Quo Failure: Municipal repair tankers take **8 to 18 hours** to mobilize. Vulnerable facilities run out of reserves in **under 120 minutes**.

THE SOLUTION

From Passive Ticket-Logging to Active Interception

Orchestrating neighborhood-level boreholes, JoJo tanks, and mobile bowsers in real-time.

TRADITIONAL CIVIC TICKETING

- ✕ **Reactive:** Merely logs what already broke after widespread citizen complaints.
- ✕ **Siloed:** Treats every complaint as an isolated ticket with zero spatial context.
- ✕ **Bottlenecked:** 100% reliant on scarce municipal repair crews and water bowsers.
- ✕ **Passive:** Sends generic SMS updates while institutions run dry.

THE RIPPLEREADY PARADIGM

- ✓ **Predictive:** Computes downstream cascade risks before buffer runs out.
- ✓ **Interdependent:** Maps crèches, clinics, and FNB merchants by urgency runway.
- ✓ **Horizontal:** Unlocks private community boreholes, church JoJo tanks, and bakkies.
- ✓ **Actionable Rails:** Real-time token handshake + automated FNB Instant Pay fuel stipends.

SYSTEM ARCHITECTURE

The 4-Stage Autonomous Engine

Deterministic, sub-second execution with end-to-end cryptographic accountability.

01

Verified Ingestion

Ward leader reports disruption with geocoordinates. The engine enforces boundaries and dedupes confirmations automatically.

`Epic 1 • POST /api/incidents`

02

Ripple Modeling

Evaluates 2.0 km perimeter. Sorts institutions by runway buffer into Tier 1 (Life), Tier 2 (Food), and Tier 3 (SME).

`Epic 2 • 3ms Spatial Math`

03

Mutual Aid Match

Matches nearest capacity within 3.5 km. Atomically locks volume, decrements reserve, and issues a 6-char OTP token.

`Epic 3 • Atomic Locking`

04

FNB Handshake

Recipient verifies physical delivery with token. Node turns GREEN, R150 fuel stipend is disbursed via FNB rails, SHA-256 logged.

`Epics 4 & 5 • Instant Pay`

STRATEGIC FIT

Why First National Bank Wins

Protecting merchants, accelerating payment rail volume, and establishing climate leadership.

🏧 SME & Merchant Turnover Continuity

Outages force township eateries, spaza shops, and bakeries to halt operations. By prioritizing commercial water continuity, RippleReady preserves daily Speedpoint POS swipe turnover, preventing business failure and reducing merchant credit risk.

R6,200/day Saved per Laundromat • **Zero Speedpoint Downtime**

⚡ FNB Instant Pay / eWallet Micro-Rails

Enables instant, transparent R150 fuel disbursements directly to volunteer bakkie drivers upon recipient token confirmation. Eliminates cumbersome municipal paperwork and provides corporate disaster funds with a fraud-proof delivery rail.

Sub-Second Payout • **Single-Use Replay-Proof Tokens**

🔐 Cryptographic Audit Ledger (SHA-256)

Every state change—from initial rupture to water delivery and payment—calculates a SHA-256 hash chained to the prior block. Guaranteed proof of delivery for ESG reporting, donor assurance, and internal compliance.

Immutable Block Hashes • **Full Traceability**

📊 Climate Resilience Indexing (CRI)

Provides FNB Commercial Property Finance and Insurance divisions with ground-truth hyper-local vulnerability metrics, enabling accurate risk pricing and preferential interest rates for climate-resilient commercial properties.

Proprietary Risk Data • **Green Financing Synergy**

ENGINEERING RIGOR

Enterprise-Grade Java & Test Coverage

Built to FNB enterprise architectural standards with 100% passing automated test suite.

BACKEND ARCHITECTURE

- **Java 25 LTS & Spring Boot 3.3.5**
- **Embedded SQLite 3 (WAL Journaling)**
- REST API with OpenAPI/Swagger UI
- Deterministic Haversine spatial math
- Sub-3ms cascade evaluation benchmark
- Single JAR executable deployment

AUTOMATED TEST SUITE

72 Tests

100% Passing in ~3.8 Seconds

✓ Epic 1: Coordinate bounds & deduplication
✓ Epic 2: Priority tiering & 100-run bench
✓ Epic 3: Over-allocation & negative units
✓ Epic 4: OTP tokens & replay attack guards
✓ Epic 5: Cryptographic hash ledger chaining
✓ SQLite: ACID persistence & WAL recovery
✓ Concurrency: 8 parallel thread safety

FRONTEND & GIS

- **React 18 + Vite + Tailwind CSS**
- Leaflet GIS with Zero Watermarks
- Dark Ops, Satellite & Streets basemaps
- Stacking context isolation (`isolate z-0`)

- Offline fallback zero-demo-failure engine

LIVE DEMONSTRATION

60-Second Hackathon Judge Flow

Five clicks showing the full journey from pipe burst to verified community resolution.

Step 1
Load Outage
2.0 km perimeter renders over Ward 18, George.

Step 2
Cascade Risks
Little Stars Crèche flagged with 2h runway.

Step 3
Pledge Aid
Locks 500L from church JoJo & generates token.

Step 4
Handshake
Recipient verifies token → Confetti fires & Node turns GREEN.

Step 5
Audit Ledger
R150 FNB stipend recorded with SHA-256 block hash.

⚡ Launch Live Demo In Operations Console →

SCALE & VISION

Impact & Strategic Roadmap

From a local Ward 18 pilot to a national resilience network embedded in the FNB App.

WARD 18 PILOT IMPACT

403
Lives Protected

R25,200
Daily SME Revenue Saved

18.4m
Avg Response Time

100%
Audit Traceability

ROLLOUT HORIZONS

- `Q1 2027` **Municipal Ward Scaling:** Expand to all 27 wards of George and Mossel Bay local municipalities.
- `Q2 2027` **IoT & Telemetry Integration:** Ingest telemetry from Western Cape Disaster Management smart meters.
- `Q3 2027` **FNB App Integration:** Native embedded feature inside the FNB Mobile App under "Community Resilience & Aid".

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FNB App of the Year Hackathon 2026 • Questions & Answers